

IRA WOLFSON

Senior Lecturer, Department of Electrical Engineering · Braude Academic College of Engineering

Quantum Researcher, Quantum X Labs (QXL) · Tel Aviv, Israel

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EDUCATION

Ph.D. in Physics (Cosmology)

Ben-Gurion University of the Negev, Israel, 2015–2019 · Advisor: Prof. Ram Brustein

Dissertation: Numerical Analysis of the Primordial Power Spectrum for Inflationary Potentials

CURRENT POSITIONS

Senior Lecturer — Department of Electrical Engineering

Braude Academic College of Engineering, Karmiel, Israel

Teaching: Electromagnetic Fields, Transmission Lines, MATLAB Programming

Quantum Researcher — Quantum X Labs (QXL · NASDAQ: QXL)

Tel Aviv, Israel, 2026–present

Quantum security and error correction; scalable quantum computing.

PREVIOUS POSITIONS

Postdoctoral Research Associate

International School for Advanced Studies (SISSA), Trieste, Italy · 2020–2022 · Supervisor: Prof. Roberto Trotta

Focus: Data Science, Machine Learning applications in Cosmology

Postdoctoral Research Fellow

Max Planck Institute for Astrophysics, Garching, Germany

Collaborations with Prof. Eiichiro Komatsu on axion-SU(2) inflation models

RESEARCH INTERESTS

- **Quantum Information:** quantum error correction, quantum sensing, quantum security, fault-tolerant quantum computing
- **Theoretical Physics:** thermodynamics, statistical mechanics, foundations of physics, gravity and black-hole entropy
- **Cosmology:** inflationary models, CMB observables, gravitational waves
- **Philosophy of Science:** Bayesian epistemology, demarcation criteria, AI ethics
- **Computational Methods:** high-dimensional Bayesian evidence calculation, GPU-accelerated algorithms
- **Interdisciplinary Research:** neuroscience, physics education

PUBLICATIONS

Peer-Reviewed Journal Articles

1. Wolfson, I. (2026). *On a geometric interpretation of the $1/4$ factor in black hole entropy*. Classical and Quantum Gravity, 43(13), 135009. DOI: 10.1088/1361-6382/ae7fe8. **NEW**

2. Wolfson, I. (2026). *Informed consent for AI consciousness research: a Talmudic framework for graduated protections*. *AI and Ethics*, 6(1), 20.
3. Ben Abu, Y., & Wolfson, I. (2024). *Short-term plasticity as 'energetic memory' of ion channel components of action potential*. *Royal Society Open Science*, 11(6), 231420.
4. Wolfson, I. (2022). *Analytic correlation of inflationary potential to power spectrum shape: limits of validity, and 'no-go' for small field model analytics*. *Journal of Cosmology and Astroparticle Physics*, 2022(01), 036.
5. Wolfson, I., Maleknejad, A., Murata, T., Komatsu, E., & Kobayashi, T. (2021). *The isotropic attractor solution of axion-SU(2) inflation: universal isotropization in Bianchi type-I geometry*. *Journal of Cosmology and Astroparticle Physics*, 2021(09), 031.
6. Wolfson, I., Maleknejad, A., & Komatsu, E. (2020). *How attractive is the isotropic attractor solution of axion-SU(2) inflation?* *Journal of Cosmology and Astroparticle Physics*, 2020(09), 047.
7. Wolfson, I., & Brustein, R. (2019). *Small field models of inflation that predict a tensor-to-scalar ratio $r=0.03$* . *Physical Review D*, 100(4), 043522.
8. Wolfson, I., & Brustein, R. (2019). *Likelihood analysis of small field polynomial models of inflation yielding a high Tensor-to-Scalar ratio*. *PLoS One*, 14(4), e0215287.
9. Ben-Dayan, I., Keating, B., Leon, D., & Wolfson, I. (2019). *Constraints on scalar and tensor spectra from N_{eff}* . *Journal of Cosmology and Astroparticle Physics*, 2019(06), 007.
10. Ben-Abu, Y., Yizhaq, H., Eshach, H., & Wolfson, I. (2019). *Interweaving the numerical kinematic symmetry principles in school and introductory university physics courses*. *Symmetry*, 11(2), 148.
11. Wolfson, I., Schramm, N.R., Biton, Y.Y., & Ben-Abu, Y. (2019). *The rain stick, a simple model for the dynamics of particles passing obstacles in a gravitational field*. *Physica A: Statistical Mechanics and its Applications*, 528, 121473.
12. Wolfson, I., & Brustein, R. (2018). *Small field models with gravitational wave signature supported by CMB data*. *PLoS One*, 13(5), e0197735.
13. Ben-Abu, Y., Wolfson, I., Eshach, H., & Yizhaq, H. (2018). *Energy, Christiaan Huygens, and the wonderful cycloid—theory versus experiment*. *Symmetry*, 10(4), 111.
14. Ben-Abu, Y., Wolfson, I., & Yizhaq, H. (2018). *Finding the speed of a bicycle in circular motion by measuring the lean angle of the bicycle*. *Physics Education*, 53(3), 035004.
15. Abu, Y.B., Wolfson, I., Bran, G., & Yizhaq, H. (2017). *Downhill cycling symmetry breaking: how the rider foils experiment*. *Physics Education*, 52(6), 065005.

Manuscripts Under Review

Physics

- Wolfson, I. *Horizon Thermality from Pure Geometry*. Under review, *Annals of Physics* (Elsevier).
- Wolfson, I. *Operational measurement of relativistic equilibrium from stochastic fields alone*. Under review, *SciPost Physics*. (arXiv:2602.16765)
- Wolfson, I. *Quantum Inaccessibility*. Under review, *Scientific Reports*. (arXiv:2511.03843)
- Wolfson, I. *Boltzmann Brain-Death*. Under review, *Foundations of Physics* (Springer).
- Wolfson, I. *One Nat per Cell: Null Geometry, Cayley Trees, and the Bekenstein-Hawking Entropy*. Under review, *EPL* (Europhysics Letters).

Computation

- Wolfson, I. *SunBURST: Deterministic GPU-Accelerated Bayesian Evidence via Mode-Centric Laplace Integration*. Under review, *Statistics and Computing* (Springer). (arXiv:2601.19957)
- Wolfson, I. *ChiSao: A GPU-Native Parallel Optimizer for Multimodal Black-Box Functions via Convergence-Anticonvergence Oscillation*. Under review, *Swarm and Evolutionary Computation* (Elsevier). (arXiv:2606.26164)

- Wolfson, I. *Knowledge and Morality Without Certainty: A Bayesian Framework*. Under review, Philosophical Studies.
- Wolfson, I. *Abortion Ethics: A Bayesian Framework for Graduated Moral Status*. Under review, Journal of Bioethical Inquiry.
- Wolfson, I. *On Scientific Demarcation*. Under review, Erkenntnis (Springer).
- Wolfson, I. *The Physical Fork of Grue*. Under review, The Philosophical Quarterly.

Software & Code

- **SunBURST** — GPU-accelerated Bayesian evidence calculation framework. irawolfson.com/sunburst; PyPI: *sunburst-bayes*.
- **ChiSao** — GPU-native multimodal black-box optimizer. PyPI: *chisao*; github.com/beastraban/chisao. **NEW**
- **INSANE** — INflationary potential Simulator and ANalysis Engine. Astrophysics Source Code Library, ascl: 2202.

CONFERENCE PRESENTATIONS & INVITED TALKS

Contributed & Conference Talks

- *Geometric Entropy: What Any Observer Can Know* — SigmaPhi 2026, Kolymbari, Crete, Greece, July 2026.
- *A New Intelligence Test for Artificial Intelligence* — Hebrew University of Jerusalem, Israel, 2023.
- *The inflationary potential inverse problem – a new outlook made possible by neural networks* — PRIMO Workshop 2022 (Post-graduate Researchers in Inverse Problems, Machine Learning, and Optimization), DMG at UniTS / SISSA, Trieste, Italy, 2022.
- *The Fault in Our Spectrum* — Learning the Universe: Cosmology × ML Workshop, Flatiron Institute (CCA), New York, USA, 2021.
- *No-go areas in the Axion-SU(2) Chromo-Natural model, and the spectator model solution* — Zooming in on Axions in the Early Universe, CERN, Switzerland, 2020.
- *Advances in GW-yielding small field models* — Inflation, Alternatives and Gravitational Waves, Ariel University, Israel, 2019.
- *Small field models GW in theory and practice — disproving and reviving the theory* — CMB from A to Z Summer School, École de Luminy, Corsica, France, 2017.
- *Gravitational waves from small field models — preferred potentials for $r=0.001$* — COSMO17, APC / IAP, Paris, France, 2017.
- *Small field models and Gravitational waves — a numerical precursor* — CosPA 2016, University of Sydney, Australia, 2016.

Invited Lectures & Colloquia

- *Data-Oriented Theoretical Cosmology* — Colloquium, NRCN, Israel, 2022.
- *Data-Oriented Theoretical Cosmology* — Colloquium, Van Swinderen Institute, University of Groningen, Netherlands, 2022.
- *Hands-on Computational Physics — the Drude Model* — Seminar, Van Swinderen Institute, University of Groningen, Netherlands, 2022.
- *Analytic Correlation of Inflationary Potential to Power Spectrum Shape* — Institutional Seminar, Max Planck Institute for Astrophysics, Munich, Germany, 2021.

Conference Organization

- Organizer — MPA/MPE Cosmology Journal Club, Max Planck Institute for Astrophysics, Munich, Germany, 2019–2020.

TEACHING EXPERIENCE

Braude Academic College of Engineering (Current)

- Electromagnetic Fields and Waves
- Transmission Lines Theory
- MATLAB Programming for Engineers

Teaching Load: 12 hours per week

TECHNICAL SKILLS

- **Programming Languages:** Python, MATLAB, C++, JavaScript, Fortran
- **Computational Tools:** GPU computing (CUDA), NumPy, SciPy, TensorFlow, JAX
- **Quantum Computing:** quantum error correction, quantum algorithms
- **Numerical Methods:** Bayesian inference, nested sampling, Monte Carlo simulations, optimization algorithms
- **Data Analysis:** machine learning, statistical analysis, high-dimensional data processing

PROFESSIONAL SERVICE

- **Peer Review:** multiple journals including JCAP, Physical Review, and AI & Ethics
- **Industry:** Quantum Researcher, Quantum X Labs (QXL)
- **Conference Participation:** regular presenter at physics, cosmology, and philosophy conferences

LANGUAGES

- Hebrew (Native)
- English (Native)

REFERENCES

Available upon request

Last Updated: July 2026